Project Demo Planning

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures the project is presented effectively, covering all key aspects in a clear and organized manner. This document outlines the demo flow, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction	Introduce HDI — explain the four indicators (life expectancy, schooling, GNI, internet usage), the four UNDP tiers (Very High / High / Medium / Low), and the purpose of the project.	2 mins	Boggavarapu Rajeswari
2	Solution Overview	Explain the three-layer architecture: HTML frontend, Flask backend, and RandomForest ML engine. Show the folder structure: app.py, templates/, data/hdi_dataset.csv.	2 mins	Boggavarapu Rajeswari
3	Code Walkthrough	Walk through app.py: load_dataset() with FileNotFoundError guard, train_model() with feature/target split, classify_hdi() with UNDP thresholds, and the POST prediction pipeline with try-except.	3 mins	Boggavarapu Rajeswari
4	Live Demo	Run the app locally at http://127.0.0.1:5000. Submit 3 test cases: (1) Life exp 82, GNI 55000 → Very High HDI 0.93. (2) Life exp 68, GNI 8000 → Medium HDI 0.59. (3) Life exp 56, GNI 1300 → Low HDI.	3 mins	Boggavarapu Rajeswari
5	Deployment	Open the live Render URL in the browser. Run the same prediction on the deployed app. Show the GitHub repository with all project files: app.py, index.html, hdi_dataset.csv, requirements.txt.	2 mins	Boggavarapu Rajeswari
6	Future Plans	Cover current limitations and the three future phases: auto-fill from CSV, dataset expansion to 150+ countries with UNDP API, and an interactive analytics dashboard with Plotly.	1 min	Boggavarapu Rajeswari

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain what HDI is, why it matters, and how ML helps predict it faster than manual formula calculation.
2	Solution Overview	Show folder structure and explain the three-layer architecture — frontend, backend, ML engine.
3	Code Walkthrough	Highlight load_dataset(), train_model(), classify_hdi(), and the @app.route POST handler in app.py.

Step	Activity	Notes
4	Live Feature Demonstration	Fill the form with 3 test cases on screen. Show Very High (0.93), Medium (0.59), and Low results.
5	Deployment Demo	Open the live Render URL. Confirm the prediction result matches the local version.
6	Q&A; Session	Answer questions on model choice, dataset, deployment approach, and planned future enhancements.